

Literature Review: [Your Project Title]

Group [X] Members:

Member Name 1 (Roll Number)

Member Name 2 (Roll Number)

Member Name 3 (Roll Number)

Member Name 4 (Roll Number)

Course: Big Data Analytics - CS 5312

Semester: Spring 2026

Instructions (Remove before submission):

- Review at least **8-12 relevant papers** for your project
- Papers should be from top-tier venues (KDD, ICDM, CIKM, NeurIPS, ICML, VLDB, SIGMOD, etc.) published in 2020-2025
- Use proper citation format (we recommend IEEE or ACM style)
- Submit both **.tex** and **.pdf** files
- Deadline: **Wednesday, April 01, 2026 (11:55 PM)**

Abstract

Provide a brief summary (150-200 words) of your literature review. Describe the main themes covered, the scope of papers reviewed, key findings from the literature, and how this review relates to your project objectives.

1 Introduction

- What is the research area/domain of your project?
- Why is this area important in the context of big data analytics?
- What specific problem are you addressing in your project?
- What are the objectives of this literature review?
- How is this review organized?

2 Background and Preliminaries

- Define key concepts and terminology used in your domain
- What foundational techniques/algorithms are relevant to your project?
- Describe any mathematical formulations or theoretical foundations
- Include relevant equations, definitions, or formal problem statements

3 Literature Survey

3.1 Category 1: [Name Your First Theme/Category]

Organize papers by themes, approaches, or techniques. For each category:

- Summarize 2-4 papers that fall under this category
- For each paper, discuss:
 - Problem addressed
 - Proposed solution/approach
 - Key contributions
 - Datasets used
 - Reported results/performance
 - Limitations

3.2 Category 2: [Name Your Second Theme/Category]

Continue with additional categories as needed. Typical categories might include:

- Different algorithmic approaches
- Different application domains
- Different data modalities
- Classical vs. deep learning methods
- Supervised vs. unsupervised approaches

3.3 Category 3: [Additional Categories as Needed]

Add more subsections as appropriate for your literature review.

4 Comparative Analysis

4.1 Summary Table

Provide a comparison table of the reviewed approaches:

Table 1: Comparison of Reviewed Approaches

Paper/Method	Technique	Dataset	Performance	Limitations
Paper 1	Approach	Dataset(s)	Metrics	Weaknesses
Paper 2	Approach	Dataset(s)	Metrics	Weaknesses
Paper 3	Approach	Dataset(s)	Metrics	Weaknesses

4.2 Critical Analysis

- What are the common strengths across reviewed approaches?
- What are the common limitations or gaps in existing work?
- How do different approaches compare in terms of:
 - Accuracy/performance metrics
 - Scalability to large datasets
 - Computational efficiency
 - Ease of implementation
 - Generalizability
- What trends do you observe in the evolution of techniques?

5 Research Gaps and Opportunities

- What problems remain unsolved or partially addressed?
- What limitations in existing work does your project aim to address?
- How does your proposed approach differ from existing solutions?
- What novel contributions does your project offer?

6 Relevance to Your Project

- Which techniques/approaches from the literature will you adopt or adapt?
- Which baselines will you compare against?
- Which datasets mentioned in the literature are relevant to your project?
- Which evaluation metrics will you use?
- How does your project build upon or extend existing work?

7 Conclusion

- Summarize the key findings from your literature review
- Restate the identified research gaps
- Briefly describe how your project addresses these gaps
- Outline anticipated challenges based on the literature

References

References

- [1] Author1, A., Author2, B., & Author3, C. (2024). *Title of the Paper*. In Proceedings of Conference Name (pp. XX-XX). Publisher.
- [2] Author1, A., & Author2, B. (2023). *Title of Another Paper*. Journal Name, Volume(Issue), Pages.